# Todd Takala

Lead Data Scientist, Machine Learning Engineer, AI/ML and Analytics

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## Summary

Data Scientist & Machine Learning Engineer with strong leadership in machine learning and statistical modeling who generated over $1.3 billion in business value. Specializations include design of data solutions, implementation of MLOps pipelines on AWS and Snowflake, enhancing customer engagement.

## Education

**Bachelor of Science in Mechanical Engineering** - Arizona State University - Tempe, AZ 06/2003 to 05/2006

## Certification & Technical Registration

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **BerkeleyX:** The Science of Happiness at Work Professional Certificate
- **Canonical:** Ubuntu Linux Professional Certificate
- **Data Camp:** AI Fundamentals Certificate • Certified SQL Associate, Data Scientist Professional Certificate
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IBM:** Advanced Deep Learning Specialist • RAG and Agentic AI Professional Certificate • Deep Learning with PyTorch, Keras and Tensorflow
- **Komatsu North America:** Certified Technical Communicator
- **US Department of Labor:** Operating Engineer Vocational Certificate

## Professional Experience

**Lead Data Scientist** - Caterpillar Inc. - Surprise, AZ 09/2022 to Present

- Architected and developed end-to-end data products using programming skills in Python and R, including an award-winning AI platform to predict equipment repair times, resulting in an $80 million reduction in warranty costs and a CEO Award
- Collaborated with senior leadership and stakeholders to facilitate understanding of the project's broad impact and create precise user requirements, ensuring alignment with the business road map through written and verbal communication
- Used LLMs and natural language processing to find similarities in service jobs and performed feature engineering to retain important parameters, which increased model accuracy by 40%
- Conducted exploratory data analysis on the training datasets, eliminating outliers using sophisticated analytics such as isolation forest and anomaly detection techniques
- Trained AI models using Python and SageMaker, applying machine learning algorithms like neural networks, PyTorch, XGBoost, TensorFlow, and LightGBM, which improved model performance and enhanced the accuracy of equipment repair time predictions, leading to better maintenance scheduling
- Engineered production-ready MLOps pipelines on AWS and Snowflake, establishing best practices for cloud architecture. Managed project lifecycles, automated enterprise networks, and implemented monitoring strategies to assess model performance and business outcomes

**Reliability Analytics Manager** - Amazon.com - Tempe, AZ 02/2022 to 09/2022

- Applied advanced analytics, mathematics, and statistics to create a new KPI for autonomous robots used in warehouse automation through big data analysis, and wrote a white paper on statistical programming, resulting in its adoption by senior management
- Led a team of business intelligence engineers and data analysts to deploy a pipeline using Shiny R, which improved real-time communication of robot status at all global fulfillment centers, enhancing operational efficiency and data analytics capabilities
- Utilized advanced data science techniques to extract actionable insights from research to driving strategic decision-making and improving operational outcomes
- Designed and implemented solutions utilizing large databases with Redshift and PostgreSQL, supporting data-driven decision-making and enhancing business intelligence capabilities

**Data Engineering Manager** - Empire-Cat - Mesa, AZ 01/2021 to 02/2022

- Developed a time-series MLOps pipeline for AI-driven forecasting in supply chain management, integrated data using SQL, R and Python, managed a component exchange program, and collaborated with clients on durability forecasting, enhancing supply chain efficiency
- Revitalized Empire's mining machine component exchange program, optimizing the supply and availability of diesel engines, transmissions, and torque converters, resulting in the identification of stock shortages and major risks to the supply chain
- Created predictive ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to $62 million in savings within 90 days

**Reliability Engineering Manager** - Empire-Cat - Mesa, AZ 07/2016 to 01/2021

- Supervised 8 engineers and technical communicators and acted as a customer-facing expert on mining equipment. Mentored up to 6 interns per year
- Headed up the root cause failure analysis program for Empire by taking a data, facts, and evidence approach, which fed the FMEA and 6 Sigma process at Caterpillar
- Implemented a data-driven failure analysis program, boosting an aging fleet's physical availability to over 90%
- Optimized resource utilization and reduced costs by leading successful reuse and salvage decisions of machinery parts

**Lead Reliability Engineer** - Empire-Cat - Mesa, AZ 01/2012 to 07/2012

- Led initiatives to maintain and expand corporate relationships with mining clients.
- Managed testing and experiment program and provided customer insights to Caterpillar engineering teams.
- Focused on deploying large off-highway truck design improvements to enhance operational efficiency.

**Technical Communications Manager** - Empire-Cat - Mesa, AZ 12/2012 to 01/2016

- Applied the DMAIC Six Sigma methodology and computer science with complex problem-solving to identify opportunities for improving operation, manufacturing, and maintenance of equipment fleet, yielding $1.2 billion in savings over 6 years
- Led a team of 5 Technical Communicators in a client-facing role to effectively convey customer feedback to Caterpillar's engineering organizations, driving improvements in product development and customer satisfaction
- Built trust and enhanced customer experience and relationships, promoting data-based decisions that improved business outcomes
- Partnered on design of experiments to improve reliability, safety and durability
- Developed Python and MS-SQL application to track performance and outcomes of prototype designs tested in the field

**Reliability Engineer** - Empire-Cat - Mesa, AZ 01/2012 to 12/2012

- Created statistical oil analysis application to analyze heavy equipment fleets according to ASTM D 7720, which became ground truth for Empire condition monitoring program
- Provided expertise in mechanical engineering and heavy equipment, serving as a key resource for improving complex machinery systems
- Investigated the primary causes of machinery downtime at mining locations, identifying key issues and recommending solutions to improve operational efficiency

**Technical Communicator, Technical Consultant** - Road Machinery, LLC - Phoenix, AZ 01/2008 to 01/2012

- Provided customer-facing engineering support by applying reliability engineering principles, resulting in improved equipment performance and customer satisfaction
- Invoiced $2.5M in consultancy fees over 3-½ years by providing specialized consultancy of mining fleet, and life cycle management
- Mentored a team of engineers at a gold mine in central Mexico, enhancing their technical skills
- Developed total cost of ownership forecast for 18 haul trucks using MATLAB and statistical models, which resulted in a sale of $72M electric haul trucks
- Conducted investigations and authored root cause failure analysis reports for high-horsepower diesel engines, transmissions and other off-highway earth moving equipment
- Designed tools and below-the-hook lifting devices to enhance workplace safety, compliance and ease of use

**Design Engineer** - Komatsu North America - Chattanooga, TN 06/2006 to 12/2007

- Designed and refined mechanical systems and components, improving product durability through FRACAS
- Contributed to prototyping and testing, analyzing performance data, and implemented product improvements and corrective actions
- Prepared comprehensive technical documentation and communicated project updates, improving team alignment and ensuring timely completion of deliverables.
- Leveraged Pro Engineer and finite element analysis to optimize engineering designs

**Engineering Intern** - Xtek Mining Services - Tempe, AZ 01/2005 to 09/2005

- Demonstrated competence in data visualization, analysis, and advanced Excel, VBA and Access leading to improved data processing efficiency
- Conducted inspections and reverse engineering of large gears and shafts for mining shovels and draglines, ensuring operational reliability and safety
- Reverse-engineered gear geometry data and developed a gear optimization application to minimize manufacturing defects, enhancing production quality
- Drafted and detailed engineering drawings, enhancing the clarity and accuracy of design documentation

**College Student** 09/2002 to 05/2006
**Operating Engineer (Contract)** - Ryan Central Inc. - McHenry County, IL 06/2002 to 09/2002

- Operated heavy equipment at mass-dirt excavation sites, contributing to the successful development of roads, building pads, and lakes on construction projects
- Loaded scrapers in push-pull configuration, enhancing the efficiency and speed of excavation processes

**Heavy Equipment Mechanic (Contract)** - Roland Machinery Co. - Marengo, IL 04/2000 to 06/2002

- Developed strong troubleshooting skills, diagnosing and resolving mechanical issues promptly, which minimized equipment downtime and improved operational efficiency
- Conducted maintenance and repair of heavy equipment, diesel engines, and hydraulic systems to ensure optimal performance
- Skillfully repaired Komatsu machinery and Cummins diesel engines, guaranteeing peak operational efficiency and extending the lifespan of the equipment
- Executed welding repairs on booms, structures, and buckets, enhancing structural integrity and safety of the equipment
- Skillfully re-manufactured key components including engines, transmissions, final drives, and hydraulic cylinders, which improved equipment performance and reduced the need for new parts
- Proven capability in handling Class B CDL service vehicles, truck-mounted cranes, and overhead cranes, with a steadfast commitment to safety and efficiency in all operations

**Heavy Equipment Apprentice Mechanic (Contract)** - Harry W. Kuhn - West Chicago, IL 12/1997 to 04/2000

- Serviced various heavy machinery such as Caterpillar, Komatsu, Bomag, Ingersoll-Rand, and Gradall, along with proficiency in repairing diesel engines
- Developed expertise in the safe loading and unloading of heavy machinery on transport trailers, ensuring secure transit and reliable operations

## Courses

- **Amazon Machine Learning University:** Tabular Data with Autogluon
- **ASME:** Y14.5M Geometric Dimensioning and Tolerancing
- **Caterpillar:** Applied Failure Analysis • SQL Server Toolset & Services • Business Tools for Service Managers • Improving Component Durability
- **Data Camp:** Bayesian Modeling in RJAGS
- **DDI:** Communicating for Leadership Success • Coaching for Peak Performance • Delegating with Purpose • Setting Performance Expectations • Resolving Workplace Conflict
- **DeepLearning.AI:** Building Systems with the ChatGPT API • ChatGPT Prompt Engineering for Developers • Finetuning Large Language Models
- **Franklin Covey:** Leadership: Great leaders - Great Teams - Great Results
- **Harper College:** Pneumatics and Hydraulics
- **IBM:** Advanced RAG with Vector Databases and Retrievers • Agentic AI with LangChain and LangGraph • Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI • Build RAG Applications • Deep Learning with Keras and Tensorflow • Deep Learning & Neural Networks with Keras • Vector Databases for RAG • Deep Learning with PyTorch
- **LinkenIn Learning:** Advanced Snowflake: Deep Dive Cloud Data Warehousing and Analytics • Building Applications Using Amazon Bedrock • Data Engineering with dbt • Designing Machine Learning Systems in the Cloud • Learning Kubernetes
- **Noria:** Oil Analysis I, II, III
- **O'Reilly Media:** Designing Machine Learning Systems in the Cloud
- **SAE:** Principles of Cost and Finance for Engineers

## Hard Skills

A/B Testing, Agentic AI, Agile Methodology, AI/ML, Anomaly Detection, Amazon Bedrock, Artificial Intelligence, AWS, BASH, CI/CD, Computer Science, RDBMS, Database Administrator, Data Science, DBT, Deep Learning, Design of Experiments, DevOps, Docker, Embedding, ETL, Fine Tuning, Advanced Excel, Financial Forecasting, Generative AI, Git, Linear Algebra, Large Language Models, LLM, Mathematics, Natural Language Processing, NLP, Pandas, Power BI, Public Cloud, Python, PyTorch, R, RAG, Research, Sagemaker, Scikit-learn, Sagemaker, Scripting Language, SDLC, Simulation, SnapLogic, Snowflake, SQL Queries, Statistical Analysis, Streamlit, SQL, Tableau, TensorFlow, Transact SQL, Transformers, Version Control

## Soft Skills

Adaptability, Attention to Detail, Business Requirements, Coaching, Communication Skills, Continuous Improvement, Creative Problem Solving, Data Driven, Decision making, Growth Mindset, Interviewing, Leadership Experience, Life Long Learning, Listening, Mentoring, Methodical, Multicultural Collaboration, Open minded, Problem Solving, Professional Services, Presentations, Public Speaking, Strategic Thinking, Team Building, Thought Leadership, Willing to Learn